

MA1513 Weekly Self Practice Set 6

1. Consider the system of differential equations $\mathbf{y}' = \begin{pmatrix} -2 & 0 \\ 5 & -2 \end{pmatrix} \mathbf{y}$.
 - (a) Show that all the eigenvectors of $\mathbf{A} = \begin{pmatrix} -2 & 0 \\ 5 & -2 \end{pmatrix}$ have the form $\begin{pmatrix} 0 \\ r \end{pmatrix}$.
 - (b) Find a generalised eigenvector of $\mathbf{A}$ that corresponds to the eigenvector $\mathbf{v} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.
 - (c) Write down a fundamental set of solutions of the SDE.
 - (d) State the type of equilibrium point of the system and its stability.
2. Consider the system of differential equations $\mathbf{y}' = \begin{pmatrix} 3 & 7 \\ -7 & 3 \end{pmatrix} \mathbf{y}$.
 - (a) Find the general (real) solution of the system. (You may use MATLAB to find the eigenvalues and eigenvectors of the underlying matrix.)
 - (b) State the type of equilibrium point of the system and its stability.
 - (c) Is the trajectory moving in clockwise or anticlockwise direction? How do you tell?